Summary in Graph

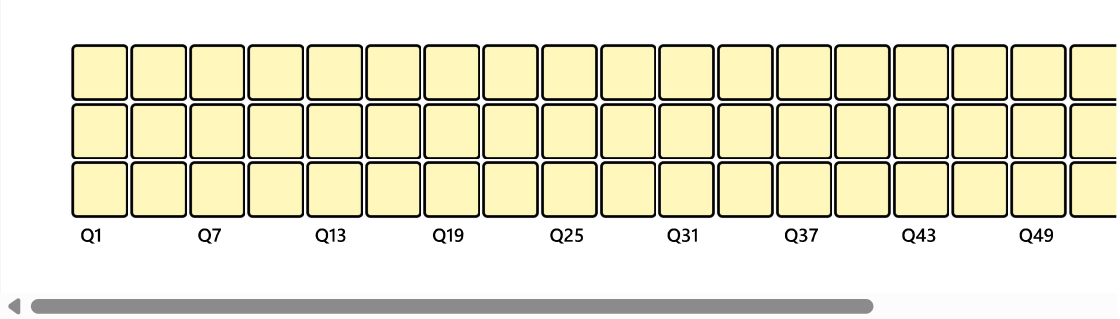
Exam Summary (GO Classes DA 2026 | All India Mock Test 7)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



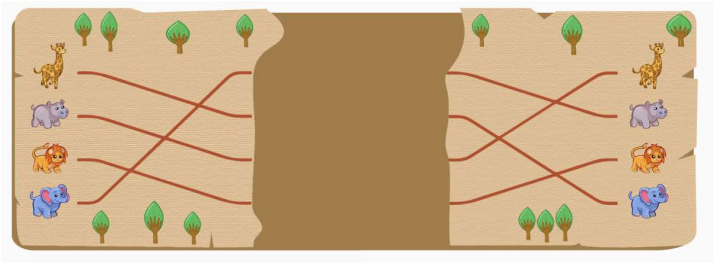
* Average Stats based on first-attempt stats of Top 10% (19) out of 193 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

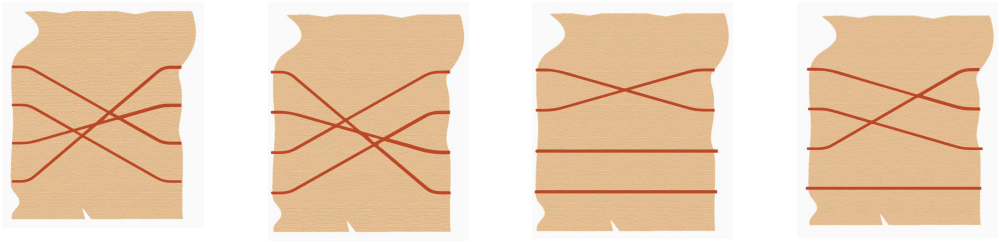
Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Four animals travelled across a plain. The following map shows their routes with a path from where each animal started to where they finished. However, the middle section of the map is missing.



Which of these pieces could be the missing middle section of the map?

A. B. C. D.



Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

If $k^2 = m^2$, which of the following must be true?

A. $k = m$

B. $k = -m$

C. $k = |m|$

D. $|k| = |m|$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

On the line segment WZ, points X and Y lie between W and Z in the order W, X, Y, Z.



If the length of segment WY is 21 units, the length of segment XZ is 26 units, and the length of segment YZ is twice the length of segment WX, what is the length of segment XY?

A. 5

B. 10

C. 11

D. 16

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Casey went on a road trip that covered 100 miles, stopping only for a lunch break along the way. The trip took 3 hours in total and her average speed while driving was 40 miles per hour. In minutes, how long was the lunch break?

A. 15

B. 30

C. 40

D. 45

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 04sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Despite the new medicine's _____ in treating diabetes, it is not _____ widely.

A. effectiveness --- prescribed

B. availability --- used

C. prescription --- available

D. acceptance --- proscribed

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.

A. 39

B. 57

C. 66

D. 93

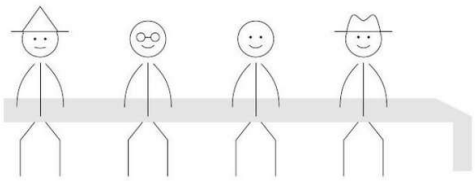
Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Four distinct students are seated in a single straight row. Each student chats with the person (or people) sitting immediately next to them.

After chatting, the students rearrange themselves in the same row such that no student is seated next to anyone they were adjacent to in the original seating.

In how many different ways can the students be rearranged to satisfy this condition?



A. 2
B. 4
C. 9
D. 12

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

A. 3.00

B. 4.99

C. 6.02

D. 7.01

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The binary operation $\square$ is defined as $a \square b = ab + (a + b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x = a$, for any a, is ____.

A. 0

B. 1

C. 2

D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

If $0 < st < 1$, then which of the following CAN be true?

A. $s < -1$ and $t > 0$

B. $s < -1$ and $t < -1$

C. $s > -1$ and $t < -1$

D. $s > 1$ and $t < -1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

X is a uniformly distributed random variable that takes values between 0 and 1. The value of $E\left\{X^3\right\}$ will be

A. 0

B. $1/8$

C. $1/4$

D. $1/2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

If the rank of a (5×6) matrix Q is 4, then which of the following statement(s) is/are correct?

A. Q will have four linearly independent rows and four linearly independent columns

B. Q will have four linearly independent rows and five linearly independent columns

C. QQ^T will be invertible

D. Q^TQ will be invertible

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

$[\mathrm{A}]$ is a square matrix which is neither symmetric nor skew-symmetric and $[\mathrm{A}]^{\mathrm{T}}$ is its transpose. The sum and difference of these matrices are defined as $[\mathrm{S}] = [\mathrm{A}] + [\mathrm{A}]^{\mathrm{T}}$ and $[\mathrm{D}] = [\mathrm{A}] - [\mathrm{A}]^{\mathrm{T}}$, respectively. Which of the following statements is **TRUE**?

A. Both $[\mathrm{S}]$ and $[\mathrm{D}]$ are symmetric

B. Both $[\mathrm{S}]$ and $[\mathrm{D}]$ are skew-symmetric

C. $[\mathrm{S}]$ is skew-symmetric and $[\mathrm{D}]$ is symmetric

D. $[\mathrm{S}]$ is symmetric and $[\mathrm{D}]$ is skew-symmetric

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

In the Taylor series expansion of e^x about $x = 2$, the coefficient of $(x-2)^4$ is

A. $1/4!$

B. $2^4/4!$

C. $e^2/4!$

D. $e^4/4!$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If $f(x)$ is an even function and a is a positive real number, then $\int_{-a}^{a} f(x)\mathrm{d}x$ equals

A. 0

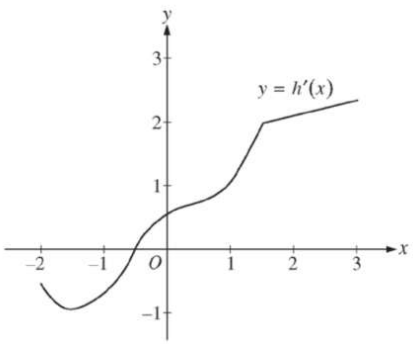
B. a

C. $2a$

D. $2\int_{0}^{a} f(x)\mathrm{d}x$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus



The graph of the derivative h' is shown above, where h is a real-valued function. Which of the following open intervals contains a value c for which the point $(c, h(c))$ is an inflection point of h?

A. $(-2, -1)$

B. $(-1, 0)$

C. $(0, 1)$

D. $(1, 2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A discrete random variable X takes values from 1 to 5 with probabilities as shown in the table. A student calculates the mean of X as 3.5 and her teacher calculates the variance of X as 1.5. Which of the following statements is true ?

k	1	2	3	4	5
$P(X = k)$	0.1	0.2	0.4	0.2	0.1

A. Both the student and the teacher are right

B. Both the student and the teacher are wrong

C. The student is wrong but the teacher is right

D. The student is right but the teacher is wrong

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A fair coin is tossed independently four times. Let X denote the number of heads obtained in the four tosses.

The probability that $X > 2$ is:

A. $1/16$

B. $1/8$

C. $1/4$

D. $5/16$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let X, Y be random variables with joint density function

$$f(x,y) = \begin{cases} cxy, & 0 \le x \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

where $c > 0$ is a constant.

Find the value of c.

Your Answer: | Correct Answer: 0.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Suppose X, Y, Z are random variables such that $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,Z) = \mathrm{Cov}(X,Z) = 0$, $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[Z] = 0$, and $\mathrm{Var}(X) = \mathrm{Var}(Y) = \mathrm{Var}(Z) = 1$. Find the value of $\mathrm{Cov}(X+Y,\ Z-Y)$.

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset E with a binary class label C (positive / negative). A binary attribute A splits the dataset E into two non-empty subsets E_1 and E_2 such that:

- E_1 contains $p_1 > 0$ positive examples and $n_1 > 0$ negative examples
- E_2 contains $p_2 > 0$ positive examples and $n_2 > 0$ negative examples

Let $I(C; A)$ denote the information gain of attribute A with respect to the class label C, defined as:

$$I(C; A) = H(C) - H(C \mid A)$$

where $H(C)$ is the entropy of the class label C, and $H(C \mid A)$ is the conditional entropy of C given A.

Which of the following statements is true?

A. $I(C; A)$ must be strictly positive

B. $I(C; A)$ must be negative

C. $I(C; A)$ can be zero

D. $I(C; A)$ is undefined

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a linear SVM trained with n labeled points in $\mathcal{R}^2$ without slack penalties and resulting in $k = 2$ support vectors ($k < n$). By adding one additional labeled training point and retraining the SVM classifier, what is the maximum number of support vectors in the resulting solution?

A. k

B. $k + 1$

C. $k + 2$

D. $n + 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider a mean-centered dataset from the banking domain consisting of 100 data points, each described by 7 features. The dataset is represented by a matrix $\mathbf{X} \in \mathbb{R}^{100\times 7}$. Principal Component Analysis (PCA) is applied to this dataset, and it is observed that the reconstruction error becomes zero after k principal components.

Later, a domain expert provides the following information. Let $\mathbf{c}_i$ denote the ith column of $\mathbf{X}$:

- The set of vectors $\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4\}$ is linearly independent.
- The following relations hold:
  - $\mathbf{c}_5 = \mathbf{c}_1 + \mathbf{c}_3$
  - $\mathbf{c}_6 = 2\mathbf{c}_3 - 3\mathbf{c}_4$
  - $\mathbf{c}_7 = \mathbf{c}_2 + 3\mathbf{c}_4$

What is the value of k? Assume that the dataset is already mean-centered.

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider two ways of representing n datapoints that belong to $\mathbb{R}^d$ in the form of a matrix:

Approach-1: A matrix $\mathbf{X}_1$ of dimension $n \times d$

Approach-2: A matrix $\mathbf{X}_2$ of dimension $d \times n$

Assume that the dataset is mean-centered. Select all correct expressions for the covariance matrix.

A. $\frac{1}{n}\mathbf{X}_1^\top \mathbf{X}_1$

B. $\frac{1}{n}\mathbf{X}_2^\top \mathbf{X}_2$

C. $\frac{1}{n}\mathbf{X}_1 \mathbf{X}_1^\top$

D. $\frac{1}{n}\mathbf{X}_2 \mathbf{X}_2^\top$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following expressions describe the growth of the following function?

$$n.\log^2(n) + 3n\sqrt{n} - 5$$

Select all that apply?

A. $O(n \cdot \sqrt{n})$

B. $O\left(n \cdot \log^2(n)\right)$

C. $\Omega(\log n)$

D. $O\left(n^3\right)$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following two mutually recursive Python-like functions:

```
def f(n):
  if n == 0:
    return True
  return g(n - 1)
```

```
def g(n):
  if n == 0:
    return False
  return f(n - 1)
```

For a given integer $n > 0$, which of the following statements is correct?

A. $f(n)$ returns True if and only if n is even

B. $f(n)$ returns True if and only if n is odd

C. Both $f(n)$ and $g(n)$ return True for all n

D. Both $f(n)$ and $g(n)$ return False for all n

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider using Quicksort to sort the array $\{18, 23, 12, 7, 26, 19, 16, 24\}$ in ascending order. The first element of a section is always chosen as the pivot.

After the first partition operation is completed (and before any recursive calls are made), which of the following arrays can be obtained?

Select all correct options.

A. $\{12, 7, 16, 18, 19, 23, 24, 26\}$

B. $\{16, 12, 7, 18, 19, 24, 23, 26\}$

C. $\{7, 12, 16, 18, 26, 19, 23, 24\}$

D. $\{12, 7, 16, 19, 18, 23, 24, 26\}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider a perfect binary tree having 15 nodes, stored in an array $A[1 \ldots 15]$ using 1-based indexing.

It is stored in such a way that during an in-order traversal (left subtree $\rightarrow$ root $\rightarrow$ right subtree) of the tree, the nodes are visited in the exact order $A[1], A[2], \ldots, A[15]$. That is, the node stored at array index i appears as the i-th node in the in-order traversal of the tree.

In other words, the first node visited in the in-order traversal is stored at array position 1, the second node visited is stored at array position 2, and so on.

Now consider the node with array index $x = 12$.

Which of the following statements are correct?

A. The left child of the node at array index 12 is located at array index 10.

B. The right child of the node at array index 12 is located at array index 14.

C. The parent of the node at array index 12 is located at array index 6.

D. None of the above.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a Depth First Search (DFS) traversal of a graph. For two distinct vertices u and v:

- $\mathrm{d}(x)$ denotes the discovery time of vertex x
- $\mathrm{f}(x)$ denotes the finish time of vertex x

Which of the following cannot occur for any two vertices u and v during a DFS?

A. $\mathrm{d}(u) < \mathrm{d}(v) < \mathrm{f}(v) < \mathrm{f}(u)$

B. $\mathrm{d}(u) < \mathrm{f}(u) < \mathrm{d}(v) < \mathrm{f}(v)$

C. $\mathrm{d}(u) < \mathrm{d}(v) < \mathrm{f}(u) < \mathrm{f}(v)$

D. $\mathrm{d}(v) < \mathrm{d}(u) < \mathrm{f}(u) < \mathrm{f}(v)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a Binary Search Tree (BST) in which all keys are distinct. A search operation for a key z is performed by repeatedly comparing z with the key at the current node and moving either to the left or right child accordingly.

Let the BST contain n keys and have height h (root at height 0).

Which of the following statements is always true?

A. The search operation always takes $\Theta(\log n)$ time

B. The search operation always takes $\Theta(h)$ time

C. The search operation takes $\Theta(n)$ time if and only if the BST is complete

D. The search operation takes $\Theta(1)$ time if the key z is present in the tree

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming

Consider the following Python code:

```
L = [1, 2, 3]
M = L
N = L[:]

M.append(4)
N.append(5)

print(L, M, N)
```

What is the output?

A. `[1, 2, 3] [1, 2, 3, 4] [1, 2, 3, 5]`

B. `[1, 2, 3, 4] [1, 2, 3, 4] [1, 2, 3, 5]`

C. `[1, 2, 3, 5] [1, 2, 3, 4] [1, 2, 3, 5]`

D. `[1, 2, 3, 4, 5] [1, 2, 3, 4] [1, 2, 3, 5]`

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the following statements related to sampling-based inference methods in probabilistic models.

Select all statements that are FALSE:

A. When there is no evidence, prior sampling is guaranteed to yield the exact same answer as inference by enumeration.

B. When collecting a sample during likelihood weighting, evidence variables are not sampled.

C. When collecting a sample during rejection sampling, variables can be sampled in any order.

D. Gibbs sampling is a technique for performing approximate inference, not exact inference.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Codd's rule of physical data independence is that

A. all information in the database is to be represented in one and only one way, namely by values in column positions within rows of tables.

B. all views that are theoretically updatable must be updatable by the system.

C. changes that are made to the physical storage representations or access methods must not require changes be made to application programs.

D. changes that are made to tables that do not modify any of the data already stored in the tables must not require changes be made to application programs.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Table T has attributes $(\mathrm{A}, \mathrm{B}, \mathrm{C})$. A is the Primary Key.

Query:

```
SELECT T1.A
FROM T AS T1, T AS T2
WHERE T1.C = T2.C
```

If table T has 1000 tuples, and the attribute C has 1000 distinct values (all unique), what is the cardinality of the result set?

Your Answer: | Correct Answer: 1000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Assuming that a table $\mathrm{R}(\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e})$ currently contains the following data values, which of the following questions about R's set of functional dependencies is/are false?

a	b	c	d	e
1	Joseph	Joe	30	x
2	Susan	Sue	20	y
3	Fred	Frodo	100	w
4	Susan	Sue	30	x
5	Christina	Chrissie	100	w
6	Jacob	Jake	50	w

A. The functional dependency $\mathrm{e} \rightarrow \mathrm{c}$ currently holds in R.

B. The functional dependency $\mathrm{b} \rightarrow \mathrm{c}$ currently holds in R.

C. The functional dependency $\mathrm{c} \rightarrow \mathrm{b}$ currently holds in R.

D. The functional dependency $(\mathrm{a}, \mathrm{b}) \rightarrow \mathrm{d}$ currently holds in R.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a 3×3 matrix. Suppose that the eigenvalues of A are $-1, 0, 1$ with respective eigenvectors $(1, -1, 0)'$, $(1, 1, -2)'$ and $(1, 1, 1)'$. Then $6A$ equals

A. $\begin{bmatrix} -1 & 5 & 2 \\ 5 & -1 & 2 \\ 2 & 2 & 2 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 5 & 3 \\ 5 & 1 & 3 \\ 3 & 3 & 3 \end{bmatrix}$

D. $\begin{bmatrix} -3 & 9 & 0 \\ 9 & -3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let P_1 and P_2 be two projection operators on a vector space. Then

A. $P_1 + P_2$ is a projection if $P_1P_2 = P_2P_1 = 0$

B. $P_1 - P_2$ is a projection if $P_1P_2 = P_2P_1 = 0$

C. $P_1 + P_2$ is a projection

D. $P_1 - P_2$ is a projection

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The linear operation $\mathrm{L}(\mathbf{x})$ is defined by the cross product $\mathrm{L}(\mathbf{x}) = \mathbf{b} \times \mathbf{x}$, where $\mathbf{b} = [\,0 \quad 1 \quad 0\,]^\top$ and $\mathbf{x} = [\,\dot{x}_1 \quad x_2 \quad x_3\,]^\top$ are three dimensional vectors. The 3×3 matrix $\mathbf{M}$ of this operation satisfies

$$\mathrm{L}(\mathbf{x}) = \mathrm{M} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then the eigenvalues of M are

A. $0, +1, -1$

B. $1, -1, 1$

C. $i, -i, 1$

D. $i, -i, 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

For any positive numbers a and b, the matrix

$$P = \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

is

A. orthogonal

B. diagonalizable

C. nonsingular

D. of rank 1

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{m \times n}$ be a real matrix. Suppose A admits a singular value decomposition. Which of the following statements related to the properties of SVD are **necessarily true**?

A. If $A = U\Sigma V^T$, then the columns of U and V are eigenvectors of A.

B. For each singular value σ_i, the corresponding vectors satisfy $Av_i = \sigma_i u_i$, where the sets $\{u_i\}$ and $\{v_i\}$ form orthonormal bases.

C. All diagonal entries of the matrix Σ are strictly positive, regardless of the rank of A.

D. The non-zero eigenvalues of both $A^T A$ and AA^T are equal to the squares of the singular values of A.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $Q \in \mathbb{R}^{n \times n}$ and $A \in \mathbb{R}^{n \times n}$. Consider the following statements involving orthonormal vectors, orthogonal matrices, and decompositions of vector spaces.

Select all statements that are correct.

A. If the columns of a matrix Q form an orthonormal set, then multiplying any vector by Q preserves its length.

B. Let $q_1, q_2 \in \mathbb{R}^n$ be orthonormal vectors. The matrix $P = q_1 q_1^T + q_2 q_2^T$ satisfies $P^2 = P$ and represents a projection operator.

C. Suppose A is a real symmetric matrix. Then every vector $x \in \mathbb{R}^n$ can be written uniquely as $x = y + z$, where $y \in \mathcal{N}(A)$ (null space of A) and $z \in \mathcal{C}(A)$ (column space of A).

D. If Q is an orthogonal matrix, then its determinant is necessarily equal to 1.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $m \times n$ matrix. Here, $\text{col}(A)$ denotes the column space of A, $\text{row}(A)$ denotes the row space of A, and $\text{null}(A)$ denotes the null space of A. Match the following matrix expressions with their possible dimensions.

	Column 1		**Column 2**
1.	$\dim(\text{col}(A))$	$P.$	$m+n$
2.	$\dim(\text{row}(A)) + \dim(\text{null}(A))$	$Q.$	n
3.	Sum of dimensions of the four fundamental subspaces	$R.$	$0, 1, \ldots, \min(m,n)$
4.	$\dim(\text{col}(A)) + \dim(\text{row}(A))$	$S.$	$0, 2, 4, \ldots, 2\min(m,n)$

A. 1:R, 2:Q, 3:P, 4:S

B. 1:Q, 2:R, 3:S, 4:P

C. 1:R, 2:P, 3:Q, 4:S

D. 1:S, 2:Q, 3:P, 4:R

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The function $f(x)$ defined by

$$f(x) = \begin{cases} 3 - x^2, & x \leq 1, \\ 3 - x, & 1 < x \leq 2, \\ x - 1, & x > 2, \end{cases}$$

has

A. a local maxima at $x = 3$ and a local minima at $x = 0$

B. a local maxima at $x = 0$ and no local minima

C. a local maxima at $x = 0$ and a local minima at $x = 2$

D. no local maxima and a local minima at $x = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-2	1	4
1	2	-3	3	-2
2	-1	1	4	5

Selected values of a function f and its first three derivatives are indicated in the table above. What is the third-degree Taylor polynomial for f about $x = 1$?

A. $2 - 3x + (3/2)x^2 - (1/3)x^3$

B. $2 - 3(x-1) + (3/2)(x-1)^2 - (1/3)(x-1)^3$

C. $2 - 3(x-1) + (3/2)(x-1)^2 - (2/3)(x-1)^3$

D. $2 - 3(x-1) + 3(x-1)^2 - 2(x-1)^3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider two independent random variables X and Y with identical distributions. The variables X and Y take values $0, 1$ and 2 with probabilities $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{4}$ respectively. What is the conditional probability $P(X + Y = 2 \mid X - Y = 0)$?

A. 0

B. $1/16$

C. $1/6$

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X have a binomial distribution with parameters n and p, where $n = 3$.

For testing the hypothesis

$$H_0 : p = 2/3 \quad \text{against} \quad H_1 : p = 1/3,$$

let a test be defined as follows:

Reject H_0 if $X \geq 2$, and accept H_0 if $X \leq 1$.

Then the probabilities of Type I error and Type II error, respectively, are:

A. $20/27$ and $20/27$

B. $7/27$ and $20/27$

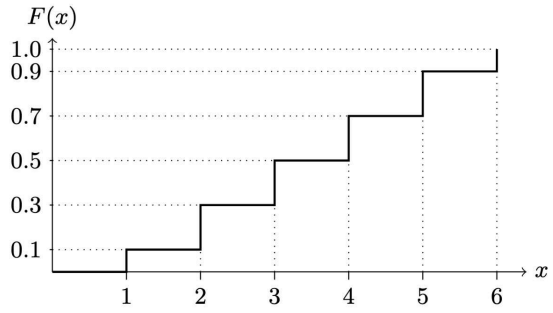
C. $20/27$ and $7/27$

D. $7/27$ and $7/27$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a biased die experiment, the random variable x of the outcome has the (cumulative) distribution function $F(x)$ as shown below.



The variance of x is

A. 1.5

B. 2.25

C. 3.5

D. 4.25

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The box 1 contains chips numbered $3, 6, 9, 12$ and 15. The box 2 contains chips numbered $6, 11, 16, 21$ and 26. Two chips, one from each box, are drawn at random. The numbers written on these chips are multiplied. The probability for the product to be an even number is

A. $6/25$

B. $2/5$

C. $3/5$

D. $19/25$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A person on a trip has a choice between a private car and public transport. The probability of using a private car is 0.45. While using public transport, the further choice available are bus and metro. Out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (rounded up to two decimals) of using a car, bus and metro respectively would be

a. $0.45, 0.30$ and 0.25

b. $0.45, 0.25$ and 0.30

c. $0.45, 0.55$ and 0

d. $0.45, 0.35$ and 0.20

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let X and Y be *independent Bernoulli random variables* such that

$$P(X = 1) = \frac{1}{2}, \qquad P(Y = 1) = p,\ 0 < p < 1$$

Define a new random variable $Z = X \oplus Y$, where the XOR operation $\oplus$ is defined as $X \oplus Y = 1$ if $X \neq Y$, and $X \oplus Y = 0$ if $X = Y$.

Select **all** correct statements.

A. Z and Y are independent for all values of p.

B. Z and X are independent for all values of p.

C. $P(Z = 1) = \frac{1}{2}$ for all values of p.

D. $P(Z = 1 \mid X = 1) = P(Z = 1 \mid X = 0)$ when $p = \frac{1}{2}$.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose you are using a Naive Bayes classifier to classify handwritten digits. Each image is represented using a single feature X, which denotes the number of black pixels in the image.

The class label is $Y \in \{0, 1, \ldots, 9\}$. For a new test image, the observed feature value is $X = 5$.

The class-conditional probabilities are given as:

$$P(X = 5 \mid Y = 0) = 0.1$$

$$P(X = 5 \mid Y = 1) = 0.01$$

$$P(X = 5 \mid Y = 2) = 0.001$$
$$\vdots$$
$$P(X = 5 \mid Y = y) = 10^{-(y+1)}$$

The prior probability of class $Y = 0$ is:

$$P(Y = 0) = 0.01$$

What is the minimum value of $P(Y = 1)$ such that the Naive Bayes classifier predicts the label $Y = 1$, assuming that in case of a tie in posterior scores, the classifier chooses class $Y = 1$?

Your Answer: | Correct Answer: 0.1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the objective function minimized in K-means clustering:

$$\sum_{j=1}^{K} \sum_{x_t \in C_j} \|x_t - r_j\|^2 \quad \text{where} \quad r_j = \frac{1}{|C_j|} \sum_{x_t \in C_j} x_t$$

Which of the following statements about the K-means algorithm is correct?

A. The K-means algorithm always finds the global minimum of the objective function.

B. The K-means algorithm may increase the objective value in some iterations.

C. The K-means algorithm is guaranteed to not increase the objective value but may converge to a local minimum.

D. The K-means algorithm always converges to the same clustering, regardless of initialization.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a binary classification problem with data points in $\mathbb{R}^2$: $x_1 = (1, 0), \quad x_2 = (0, 1), \quad x_3 = (-1, 0)$ with corresponding labels $y_1 = +1, \quad y_2 = +1, \quad y_3 = -1$ A linear classifier of the form $f(x) = w_0 + w_1 x_1 + w_2 x_2$ is trained using the perceptron learning algorithm, where the initial weight vector is $w = (0, 0, 0)^T$, points lying on the decision boundary are treated as misclassified, and the training order is $x_1 \to x_2 \to x_3 \to x_1 \to \cdots$. The algorithm terminates when all points are correctly classified. What is the final value of the bias term w_0?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider 3 data points $(1, 1), (1, 0), (0, 0)$ with labels $0, 1, 0$, respectively. Two models are trained using **logistic regression:**

1. **Model 1** : $\sigma(w_1 x_1 + w_2 x_2)$,
2. **Model 2** : $\sigma(w_0 + w_1 x_1 + w_2 x_2)$,
   where $\sigma(z)$ is the sigmoid function: $\frac{1}{1+e^{-z}}$.

Now, suppose the label of $(0, 0)$ is changed from 0 to 1.

What happens to the learned parameters (w) for Model 1 and Model 2?

A. The parameters for both Model 1 and Model 2 change.

B. The parameters for Model 1 remain the same, but the parameters for Model 2 change.

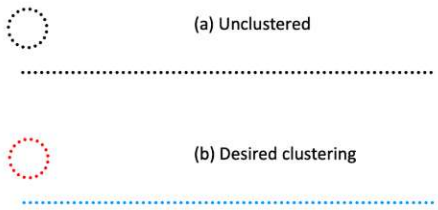
C. The parameters for Model 1 change, but the parameters for Model 2 remain the same.

D. The parameters for both Model 1 and Model 2 remain the same.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following methods will cluster the data in panel (a) of the figure below into the two clusters (red circle and blue horizontal line) shown in panel (b)? Every dot in the circle and the line is a data point. In all the options that involve hierarchical clustering, the algorithm is run until we obtain two clusters.



A. Hierarchical agglomerative clustering with Euclidean distance and complete linkage

B. Hierarchical agglomerative clustering with Euclidean distance and single linkage

C. Hierarchical agglomerative clustering with Euclidean distance and centroid linkage

D. k-means clustering with $k = 2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let G be a graph in which all edge weights are equal to the same positive constant. Suppose breadth-first search (BFS) and Dijkstra's algorithm are both run on G from the same start vertex, and both algorithms resolve ties in the same way.

Which of the following statements is correct?

A. The order in which nodes are removed from the BFS queue is the same as the order in which nodes are removed from the priority queue in Dijkstra's algorithm.

B. BFS and Dijkstra's algorithm may produce different node removal orders even with identical tie-breaking.

C. The two algorithms produce the same order only when all edge weights are equal to 1.

D. The order depends only on the number of vertices and not on the edge weights.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

You are given the initial array $[22, 10, 14, 37, 14, 4, 3]$.

For each array listed in **Column 1**, determine whether it could appear at any point during the execution of

Insertion Sort or Selection Sort applied to the initial array. If neither algorithm could produce the given array, choose **None.**

	Column 1		Column 2
1.	$[3, 4, 14, 37, 14, 10, 22]$	$P.$	Insertion Sort
2.	$[10, 22, 3, 14, 14, 37, 4]$	$Q.$	Selection Sort
3.	$[10, 14, 14, 22, 37, 4, 3]$	$R.$	None

A. $1 : \mathrm{Q},\ 2 : \mathrm{R},\ 3 : \mathrm{P}$

B. $1 : \mathrm{P},\ 2 : \mathrm{Q},\ 3 : \mathrm{R}$

C. $1 : \mathrm{Q},\ 2 : \mathrm{P},\ 3 : \mathrm{R}$

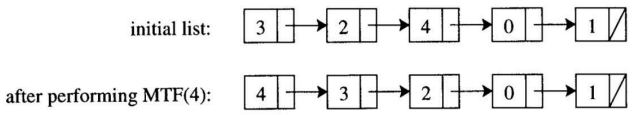
D. $1 : \mathrm{P},\ 2 : \mathrm{R},\ 3 : \mathrm{Q}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a singly linked list containing the numbers $0, 1, \ldots, n$. The operation $\mathrm{MTF}(i)$ searches the list from the beginning until i is found, and then moves the node containing i to the front of the linked list.

For example,



Suppose that the list is initially sorted in increasing order and a sequence of m operations $\mathrm{MTF}(i_1), \mathrm{MTF}(i_2), \ldots, \mathrm{MTF}(i_m)$ is performed. Let L denote the final list.

Let S be the set of elements that appear at least once in the sequence $i_1, i_2, \ldots, i_m$. For each $x \in S$, define $t(x)$ to be the largest index $r \in \{1, 2, \ldots, m\}$ such that $i_r = x$; that is, $t(x)$ denotes the time (operation number) at which x is moved to the front for the last time.

Which of the following statements about the final list L is correct?

A. The elements of S appear in L in decreasing order of $t(x)$, followed by the elements not in S in increasing order.

B. The elements of S appear in L in increasing order of $t(x)$, followed by the elements not in S in decreasing order.

C. The final list L depends only on the multiset of values in $\{i_1, \ldots, i_m\}$ and not on their order.

D. If two elements x and y satisfy $t(x) = t(y)$, then their relative order in L is the same as in the initial list.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

You are given a hash table of size M that uses linear probing for collision resolution. The table is half-full, i.e., exactly $M/2$ cells are occupied.

The occupied cells are consecutive throughout the table. The cost of inserting an item is defined as the number of probes required to find an empty cell. Assume that the initial hash position of a key is uniformly distributed over the table.

What is the average time complexity to insert a new item?

A. $\Theta(M)$

B. $\Theta(\log M)$

C. $\Theta(1)$

D. $\Theta(\sqrt{M})$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Numerical Type | Award: 2 | Penalty: 0 | Programming in Python

Consider the following Python function:

```
def func(L):
  if len(L) == 1:
    return L[0]
  return max(L[0], func(L[1:]))
```

Here, `max(x, y)` returns the larger of the two values `x` and `y`.

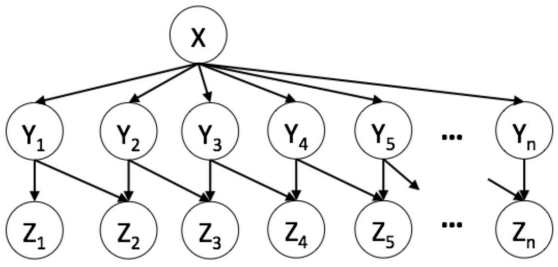
Let the list be `A = [6, 2, 9, 4, 1]`

What is the output of `print(func(A))`?

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following Bayesian Network, where all variables are binary:



All variables $Z_1, Z_2, \ldots, Z_n$ are observed as evidence. We want to compute

$$P(Y_n \mid Z_1 = z_1, Z_2 = z_2, \ldots, Z_n = z_n)$$

using variable elimination. If the first variable eliminated is Y_1, which of the following is the new factor generated?

A. $P(Z_1 \mid X)$

B. $P(Z_1, Z_2 \mid Y_2, X)$

C. $P(Y_2 \mid Z_1, Z_2, X)$

D. $P(Z_2 \mid Y_2, X)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of these following relational algebra expressions are always true ?

A. $(R \bowtie S) \bowtie T = (T \bowtie R) \bowtie S$

B. $\sigma_A(\sigma_B(R)) = \sigma_B(\sigma_A(R))$

C. $\Pi_A(\Pi_B(R)) = \Pi_B(\Pi_A(R))$

D. $\sigma_A(R \bowtie S) = \sigma_A(R) \bowtie S$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Suppose the size of a search key field is $V = 16$ bytes, the size of a data (record) pointer is $Pr = 6$ bytes, the size of a block pointer (i.e. tree pointer) is $P = 8$ bytes, and the block size is $B = 1024$ bytes.

A B^+-tree is to be constructed.

- Order of an internal node: maximum number of child pointers it can have.
- Order of a leaf node: maximum number of key–record pointer pairs it can store.

What is the absolute value of the difference between the order of an internal node and the order of a leaf node?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are true?

A. The argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion $q \rightarrow r$ is valid iff the argument form with premises $p_1, p_2, \ldots, p_n, r$, and conclusion q is valid.

B. The argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion $q \rightarrow r$ is valid iff the argument form with premises $p_1, p_2, \ldots, p_n, q$, and conclusion r is valid.

C. The argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion $q \rightarrow r$ is valid iff the argument form with premises $p_1, p_2, \ldots, p_n, q$, and conclusion $\sim r$ is valid.

D. The argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion $q \rightarrow r$ is valid iff the argument form with premises $p_1, p_2, \ldots, p_n, \sim r$, and conclusion $\sim q$ is valid.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Let $p(x), q(x)$, and $r(x)$ be the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 7x + 10 = 0 \\ q(x): &\quad x^2 - 2x - 3 = 0 \\ r(x): &\quad x < 0 \end{aligned}$$

For the universe of all positive integers, which of the following statements is/are false?

A. $\forall x[p(x) \rightarrow \neg r(x)]$

B. $\forall x[q(x) \rightarrow r(x)]$

C. $\exists x[q(x) \rightarrow r(x)]$

D. $\exists x[p(x) \rightarrow r(x)]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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